Problem 4.29: Zettili Notation Throughout

Systems-of-equations method with $|j_1, j_2;\, m_1, m_2\rangle$ product states

CG coefficients written as $\langle j_1, j_2;\, m_1, m_2 \,|\, J, M\rangle$

The Problem

Find the ratio of cross sections for:

(a) $\pi^- + p \to K^0 + \Sigma^0$

(b) $\pi^- + p \to K^+ + \Sigma^-$

(c) $\pi^+ + p \to K^+ + \Sigma^+$

First when $I = 3/2$ dominates, then when $I = 1/2$ dominates.

Zettili notation (Eq. 7.159)

A total-isospin state $|J, M\rangle$ is expanded in terms of direct-product states as:

$$|J, M\rangle = \sum_{m_1} \sum_{m_2} \langle j_1, j_2;\, m_1, m_2 \,|\, J, M\rangle \; |j_1, j_2;\, m_1, m_2\rangle \tag{7.159}$$

where:

- $|j_1, j_2;\, m_1, m_2\rangle = |j_1, m_1\rangle \otimes |j_2, m_2\rangle$ is the direct-product ket
- $\langle j_1, j_2;\, m_1, m_2 \,|\, J, M\rangle$ is the CG coefficient — read from the table
- Only terms with $m_1 + m_2 = M$ survive (all others are zero)

How to read the CG table:

- Tables are organized as $j_1 \times j_2$ (bigger j first)
- **Row** = which $|J, M\rangle$ state
- **Column** = which $|m_1, m_2\rangle$ combination
- **Entry** = a number with a **square root understood over it**
- Reading down a column gives you the decomposition of a product state
- Reading across a row gives you the decomposition of a total-J state

Reaction (a): $\pi^- + p \to K^0 + \Sigma^0$

Step 1: Write Out the Reaction

$$\pi^- + p \longrightarrow K^0 + \Sigma^0 \tag{1}$$

Work the **initial state** first, then the **final state**.

Initial State: $\pi^- + p$

Step 2: Find I and I_3 for each particle, write the kets

Pion ($\pi^- = d\bar{u}$):

- $I_3 = (-1/2) + (-1/2) = -1$
- Part of a triplet (π^+, π^0, π^-): $2I + 1 = 3 \Rightarrow I = 1$
- Also: $I_3 = -1$ only fits in $I \geq 1$, and the stretched state forces $I = 1$

$$\pi^- = |I_\pi = 1,\ I_{3,\pi} = -1\rangle \tag{2}$$

Proton ($p = uud$):

- $I_3 = +1/2 + 1/2 - 1/2 = +1/2$
- Part of a doublet (p, n): $2I + 1 = 2 \Rightarrow I = 1/2$

$$p = |I_p = \tfrac{1}{2},\ I_{3,p} = +\tfrac{1}{2}\rangle \tag{3}$$

The direct-product ket in Zettili notation (pion first, since $I_\pi = 1$ is bigger):

$$|\pi^- p\rangle = |1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle \tag{4}$$

This means: $|j_1 = 1,\ j_2 = 1/2;\ m_1 = -1,\ m_2 = +1/2\rangle$.

Step 3: Triangle rule and I_3 conservation

Triangle rule: combining $j_1 = 1$ and $j_2 = 1/2$:

$$|1 - \tfrac{1}{2}| \leq I \leq 1 + \tfrac{1}{2} \quad \Longrightarrow \quad \frac{1}{2} \leq I \leq \frac{3}{2} \tag{5}$$

So $I = 1/2$ or $I = 3/2$.
I_3 **conservation**: $M = m_1 + m_2 = (-1) + (+1/2) = -1/2$.
Check which $|J, M\rangle$ states can have $M = -1/2$:

- $|3/2, -1/2\rangle$: $I = 3/2$ allows $I_3 = -1/2$. **Yes.**
- $|1/2, -1/2\rangle$: $I = 1/2$ allows $I_3 = -1/2$. **Yes.**

We need both: $|3/2, -1/2\rangle$ and $|1/2, -1/2\rangle$.

Step 4: Decompose total-isospin kets using the CG table

Using (7.159) with $j_1 = 1$, $j_2 = 1/2$. The product states with $m_1 + m_2 = -1/2$ are:

- $|1, \frac{1}{2};\ 0, -\frac{1}{2}\rangle$ (that's $|\pi^0\rangle|n\rangle$)
- $|1, \frac{1}{2};\ -1, +\frac{1}{2}\rangle$ (that's $|\pi^-\rangle|p\rangle$)

Note: $|1, \frac{1}{2};\ -\frac{1}{2}, 0\rangle$ does NOT exist because the pion ($j_1 = 1$) can't have $m_1 = -1/2$ and the nucleon ($j_2 = 1/2$) can't have $m_2 = 0$.

Decompose $|3/2, -1/2\rangle$: Using (7.159):

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \langle 1, \tfrac{1}{2};\ 0, -\tfrac{1}{2} \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle\ |1, \tfrac{1}{2};\ 0, -\tfrac{1}{2}\rangle + \langle 1, \tfrac{1}{2};\ -1, +\tfrac{1}{2} \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle\ |1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle \tag{6}$$

Read the CG table ($1 \times 1/2$ table, row $|3/2, -1/2\rangle$):

$$\langle 1, \tfrac{1}{2};\ 0, -\tfrac{1}{2} \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle = \sqrt{\frac{2}{3}} \quad \text{(table entry: } 2/3\text{, take } \surd\text{)} \tag{7}$$

$$\langle 1, \tfrac{1}{2};\ -1, +\tfrac{1}{2} \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle = \sqrt{\frac{1}{3}} \quad \text{(table entry: } 1/3\text{, take } \surd\text{)} \tag{8}$$

So:

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\ |1, \tfrac{1}{2};\ 0, -\tfrac{1}{2}\rangle + \sqrt{\frac{1}{3}}\ |1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle \tag{Eq. 1}$$

In particle language:

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\ |\pi^0\rangle|n\rangle + \sqrt{\frac{1}{3}}\ |\pi^-\rangle|p\rangle \tag{Eq. 1′}$$

Decompose $|1/2, -1/2\rangle$: Same procedure, read row $|1/2, -1/2\rangle$:

$$\langle 1, \tfrac{1}{2};\ 0, -\tfrac{1}{2} \,|\, \tfrac{1}{2}, -\tfrac{1}{2}\rangle = \sqrt{\frac{1}{3}} \quad \text{(table entry: } 1/3\text{)} \tag{9}$$

$$\langle 1, \tfrac{1}{2};\ -1, +\tfrac{1}{2} \,|\, \tfrac{1}{2}, -\tfrac{1}{2}\rangle = -\sqrt{\frac{2}{3}} \quad \text{(table entry: } -2/3\text{)} \tag{10}$$

$$\left|\frac{1}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\ |1, \tfrac{1}{2};\ 0, -\tfrac{1}{2}\rangle - \sqrt{\frac{2}{3}}\ |1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle \tag{Eq. 2}$$

In particle language:

$$\left|\frac{1}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\ |\pi^0\rangle|n\rangle - \sqrt{\frac{2}{3}}\ |\pi^-\rangle|p\rangle \tag{Eq. 2′}$$

Quick checks on Eq. 1 and Eq. 2

Normalization:

- Eq. 1: $\left(\sqrt{2/3}\right)^2 + \left(\sqrt{1/3}\right)^2 = 2/3 + 1/3 = 1$ ✓
- Eq. 2: $\left(\sqrt{1/3}\right)^2 + \left(-\sqrt{2/3}\right)^2 = 1/3 + 2/3 = 1$ ✓

Orthogonality:

$$\sqrt{\frac{2}{3}} \cdot \sqrt{\frac{1}{3}} + \sqrt{\frac{1}{3}} \cdot \left(-\sqrt{\frac{2}{3}}\right) = \frac{\sqrt{2}}{3} - \frac{\sqrt{2}}{3} = 0 \quad \checkmark \tag{11}$$

Step 5: Invert using systems of equations to get $|\pi^- p\rangle$

We want to isolate $|1, \frac{1}{2};\ -1, +\frac{1}{2}\rangle = |\pi^- p\rangle$ from Eq. 1 and Eq. 2.
Goal: eliminate $|1, \frac{1}{2};\ 0, -\frac{1}{2}\rangle = |\pi^0 n\rangle$.

Multiply Eq. 1 by $\sqrt{1/3}$:

$$\begin{aligned}\sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle &= \sqrt{\frac{1}{3}} \cdot \sqrt{\frac{2}{3}}\,|1, \tfrac{1}{2};\ 0, -\tfrac{1}{2}\rangle + \sqrt{\frac{1}{3}} \cdot \sqrt{\frac{1}{3}}\,|1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle \\ &= \frac{\sqrt{2}}{3}\,|1, \tfrac{1}{2};\ 0, -\tfrac{1}{2}\rangle + \frac{1}{3}\,|1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle \end{aligned} \tag{1a}$$

Multiply Eq. 2 by $-\sqrt{2/3}$:

$$\begin{aligned}-\sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle &= -\sqrt{\frac{2}{3}} \cdot \sqrt{\frac{1}{3}}\,|1, \tfrac{1}{2};\ 0, -\tfrac{1}{2}\rangle + \sqrt{\frac{2}{3}} \cdot \sqrt{\frac{2}{3}}\,|1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle \\ &= -\frac{\sqrt{2}}{3}\,|1, \tfrac{1}{2};\ 0, -\tfrac{1}{2}\rangle + \frac{2}{3}\,|1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle \end{aligned} \tag{2a}$$

Add (1a) + (2a): The $|1, \frac{1}{2};\ 0, -\frac{1}{2}\rangle$ terms cancel:

$$\begin{aligned}&\sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \\ &\quad = \left(\frac{1}{3} + \frac{2}{3}\right)|1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle = 1 \cdot |1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle \end{aligned} \tag{12}$$

Therefore:

$$\boxed{|\pi^- p\rangle = |1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \tag{Initial}$$

Shortcut: reading the CG table by column

Instead of solving the system of equations, you can read the **column** $m_1 = -1, m_2 = +1/2$ of the CG table directly. Reading down:

$$\langle 1, \tfrac{1}{2};\ -1, +\tfrac{1}{2} \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle = \sqrt{1/3} \tag{13}$$

$$\langle 1, \tfrac{1}{2};\ -1, +\tfrac{1}{2} \,|\, \tfrac{1}{2}, -\tfrac{1}{2}\rangle = -\sqrt{2/3} \tag{14}$$

These are exactly the coefficients we got by solving the system! Reading by column and inverting the system are equivalent methods.

Final State: $K^0 + \Sigma^0$

Step 2: Find I and I_3 for each particle, write the kets

Kaon ($K^0 = d\bar{s}$):

- $I_3 = (-1/2) + 0 = -1/2 \quad (\bar{s}$ has $I = 0)$
- Only one light quark $\Rightarrow I = 1/2$ automatically

$$K^0 = |I_K = \tfrac{1}{2},\ I_{3,K} = -\tfrac{1}{2}\rangle \tag{15}$$

Sigma ($\Sigma^0 = uds$):

- $I_3 = (+1/2) + (-1/2) + 0 = 0$
- Triplet $(\Sigma^+, \Sigma^0, \Sigma^-)$: $2I + 1 = 3 \Rightarrow I = 1$

$$\Sigma^0 = |I_\Sigma = 1,\ I_{3,\Sigma} = 0\rangle \tag{16}$$

Direct-product ket (kaon first):

$$|K^0\Sigma^0\rangle = |\tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0\rangle \tag{17}$$

This means: $|j_1 = 1/2,\ j_2 = 1;\ m_1 = -1/2,\ m_2 = 0\rangle$.

Step 3: Triangle rule and I_3 conservation

Triangle rule: combining $j_1 = 1/2$ (kaon) and $j_2 = 1$ (sigma):

$$|1/2 - 1| \le I \le 1/2 + 1 \quad \Longrightarrow \quad \frac{1}{2} \le I \le \frac{3}{2} \tag{18}$$

So $I = 1/2$ or $I = 3/2$.
I_3 **conservation**: $M = (-1/2) + 0 = -1/2$. Both $|3/2, -1/2\rangle$ and $|1/2, -1/2\rangle$ survive.

Step 4: Decompose total-isospin kets using the CG table

Coupling order: $1/2 \times 1$ vs. the $1 \times 1/2$ table

Our product ket is $|\frac{1}{2}, 1;\ m_1, m_2\rangle$ (kaon first). The CG table is printed for $1 \times 1/2$ (sigma first). To use the table, we swap the coupling order and apply the phase:

$$\langle \tfrac{1}{2}, 1;\ m_K, m_\Sigma \,|\, J, M\rangle = (-1)^{1+1/2-J}\ \langle 1, \tfrac{1}{2};\ m_\Sigma, m_K \,|\, J, M\rangle \tag{19}$$

- $J = 3/2$: phase $= (-1)^{1+1/2-3/2} = (-1)^0 = +1$ (keep as-is)
- $J = 1/2$: phase $= (-1)^{1+1/2-1/2} = (-1)^1 = -1$ (flip sign)

Product states with $m_K + m_\Sigma = -1/2$:

- $|\frac{1}{2}, 1;\ -\frac{1}{2}, 0\rangle$ (that's $|K^0\rangle|\Sigma^0\rangle$)
- $|\frac{1}{2}, 1;\ +\frac{1}{2}, -1\rangle$ (that's $|K^+\rangle|\Sigma^-\rangle$)

Decompose $|3/2, -1/2\rangle$: Using (7.159) adapted for $1/2 \times 1$:

$$\begin{aligned}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = {} & \langle \tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0 \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle\ |\tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0\rangle \\ & + \langle \tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1 \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle\ |\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1\rangle\end{aligned} \tag{20}$$

Look up each CG coefficient. For the first one:

$$\begin{aligned}&\langle \tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0 \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle \\ &\quad = (-1)^{1+1/2-3/2} \times \langle 1, \tfrac{1}{2};\ 0, -\tfrac{1}{2} \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle \\ &\quad = (+1) \times \sqrt{\frac{2}{3}} = \sqrt{\frac{2}{3}}\end{aligned} \tag{21}$$

For the second one:

$$\begin{aligned}&\langle \tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1 \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle \\ &\quad = (-1)^{1+1/2-3/2} \times \langle 1, \tfrac{1}{2};\ -1, +\tfrac{1}{2} \,|\, \tfrac{3}{2}, -\tfrac{1}{2}\rangle \\ &\quad = (+1) \times \sqrt{\frac{1}{3}} = \sqrt{\frac{1}{3}}\end{aligned} \tag{22}$$

So:

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\ |\tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0\rangle + \sqrt{\frac{1}{3}}\ |\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1\rangle \tag{Eq. 3}$$

In particle language:

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\ |K^0\rangle|\Sigma^0\rangle + \sqrt{\frac{1}{3}}\ |K^+\rangle|\Sigma^-\rangle \tag{Eq. 3′}$$

Decompose $|1/2, -1/2\rangle$: Now $J = 1/2$, so the phase is (-1).

First CG coefficient:

$$\begin{aligned}&\langle \tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0 \,|\, \tfrac{1}{2}, -\tfrac{1}{2}\rangle \\ &\quad = (-1)^{1+1/2-1/2} \times \langle 1, \tfrac{1}{2};\ 0, -\tfrac{1}{2} \,|\, \tfrac{1}{2}, -\tfrac{1}{2}\rangle \\ &\quad = (-1) \times \sqrt{\frac{1}{3}} = -\sqrt{\frac{1}{3}}\end{aligned} \tag{23}$$

Second CG coefficient:

$$\begin{aligned}&\langle \tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1 \,|\, \tfrac{1}{2}, -\tfrac{1}{2}\rangle \\ &\quad = (-1)^{1+1/2-1/2} \times \langle 1, \tfrac{1}{2};\ -1, +\tfrac{1}{2} \,|\, \tfrac{1}{2}, -\tfrac{1}{2}\rangle \\ &\quad = (-1) \times \left(-\sqrt{\frac{2}{3}}\right) = +\sqrt{\frac{2}{3}}\end{aligned} \tag{24}$$

So:

$$\left|\frac{1}{2}, -\frac{1}{2}\right\rangle = -\sqrt{\frac{1}{3}}\, |\tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0\rangle + \sqrt{\frac{2}{3}}\, |\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1\rangle \tag{Eq. 4}$$

In particle language:

$$\left|\frac{1}{2}, -\frac{1}{2}\right\rangle = -\sqrt{\frac{1}{3}}\, |K^0\rangle|\Sigma^0\rangle + \sqrt{\frac{2}{3}}\, |K^+\rangle|\Sigma^-\rangle \tag{Eq. 4′}$$

Quick checks on Eq. 3 and Eq. 4

Normalization:

- Eq. 3: $2/3 + 1/3 = 1$ ✓
- Eq. 4: $1/3 + 2/3 = 1$ ✓

Orthogonality:

$$\sqrt{\frac{2}{3}} \cdot \left(-\sqrt{\frac{1}{3}}\right) + \sqrt{\frac{1}{3}} \cdot \sqrt{\frac{2}{3}} = -\frac{\sqrt{2}}{3} + \frac{\sqrt{2}}{3} = 0 \quad \checkmark \tag{25}$$

Step 5: Invert using systems of equations to get $|K^0\Sigma^0\rangle$

We want to isolate $|\tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0\rangle = |K^0\Sigma^0\rangle$ from Eq. 3 and Eq. 4.
Goal: eliminate $|\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1\rangle = |K^+\Sigma^-\rangle$.

Multiply Eq. 3 by $\sqrt{2/3}$:

$$\sqrt{\frac{2}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \frac{2}{3}\, |\tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0\rangle + \frac{\sqrt{2}}{3}\, |\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1\rangle \tag{3a}$$

Multiply Eq. 4 by $-\sqrt{1/3}$:

$$-\sqrt{\frac{1}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle = \frac{1}{3}\,|\tfrac{1}{2},1;\ -\tfrac{1}{2},0\rangle - \frac{\sqrt{2}}{3}\,|\tfrac{1}{2},1;\ +\tfrac{1}{2},-1\rangle \tag{4a}$$

Add (3a) + (4a): The $|\frac{1}{2},1;\ +\frac{1}{2},-1\rangle$ terms cancel:

$$\sqrt{\frac{2}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle$$
$$= \left(\frac{2}{3}+\frac{1}{3}\right)|\tfrac{1}{2},1;\ -\tfrac{1}{2},0\rangle = 1\cdot|\tfrac{1}{2},1;\ -\tfrac{1}{2},0\rangle \tag{26}$$

Therefore:

$$\boxed{|K^0\Sigma^0\rangle = |\tfrac{1}{2},1;\ -\tfrac{1}{2},0\rangle = \sqrt{\frac{2}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle} \tag{Final-a}$$

Amplitude for Reaction (a)

Step 6: Compute the amplitude

From (Initial) and (Final-a):

$$|\pi^- p\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{27}$$

$$|K^0\Sigma^0\rangle = \sqrt{\frac{2}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{28}$$

Multiply CG coefficients channel by channel:

Channel	**Initial coeff.**	**Final coeff.**	**Product**
$I = 3/2$	$\sqrt{1/3}$	$\sqrt{2/3}$	$\sqrt{1/3}\times\sqrt{2/3} = \dfrac{\sqrt{2}}{3}$
$I = 1/2$	$-\sqrt{2/3}$	$-\sqrt{1/3}$	$(-\sqrt{2/3})\times(-\sqrt{1/3}) = +\dfrac{\sqrt{2}}{3}$

$$\boxed{\mathcal{M}_a = \frac{\sqrt{2}}{3}\,\mathcal{A}_{3/2} + \frac{\sqrt{2}}{3}\,\mathcal{A}_{1/2} = \frac{\sqrt{2}}{3}\left(\mathcal{A}_{3/2}+\mathcal{A}_{1/2}\right)} \tag{29}$$

$$|\mathcal{M}_a|^2 = \frac{2}{9}\left|\mathcal{A}_{3/2}+\mathcal{A}_{1/2}\right|^2 \tag{30}$$

Reaction (b): $\pi^- + p \to K^+ + \Sigma^-$

Step 1: Write Out the Reaction

$$\pi^- + p \longrightarrow K^+ + \Sigma^- \tag{31}$$

Initial state is the same as reaction (a):

$$|\pi^- p\rangle = |1, \tfrac{1}{2};\ -1, +\tfrac{1}{2}\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \qquad \text{(Initial — same)}$$

Final State: $K^+ + \Sigma^-$

Step 2: Find I and I_3, write the kets

Kaon ($K^+ = u\bar{s}$): $I = 1/2, \quad I_3 = +1/2.$
Sigma ($\Sigma^- = dds$): $I = 1, \quad I_3 = -1.$
Direct-product ket (kaon first):

$$|K^+\Sigma^-\rangle = |\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1\rangle \tag{32}$$

I_3 check: $(+1/2) + (-1) = -1/2$. Matches the initial state. ✓

Steps 3–4: We already have the decompositions!

From the work on reaction (a), we already decomposed $|3/2, -1/2\rangle$ and $|1/2, -1/2\rangle$ in terms of $K\Sigma$ product kets (Eq. 3 and Eq. 4). Now we just need to isolate the **other** product ket: $|\frac{1}{2}, 1;\ +\frac{1}{2}, -1\rangle = |K^+\Sigma^-\rangle$.

Step 5: Invert to get $|K^+\Sigma^-\rangle$

From Eq. 3 and Eq. 4:

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\,|\tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0\rangle + \sqrt{\frac{1}{3}}\,|\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1\rangle \qquad \text{(Eq. 3)}$$

$$\left|\frac{1}{2}, -\frac{1}{2}\right\rangle = -\sqrt{\frac{1}{3}}\,|\tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0\rangle + \sqrt{\frac{2}{3}}\,|\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1\rangle \qquad \text{(Eq. 4)}$$

Goal: eliminate $|\frac{1}{2}, 1;\ -\frac{1}{2}, 0\rangle = |K^0\Sigma^0\rangle$.

Multiply Eq. 3 by $\sqrt{1/3}$:

$$\sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \frac{\sqrt{2}}{3}\,|\tfrac{1}{2}, 1;\ -\tfrac{1}{2}, 0\rangle + \frac{1}{3}\,|\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, -1\rangle \qquad \text{(3b)}$$

Multiply Eq. 4 by $\sqrt{2/3}$:

$$\sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle = -\frac{\sqrt{2}}{3}\,|\tfrac{1}{2},1;\ -\tfrac{1}{2},0\rangle + \frac{2}{3}\,|\tfrac{1}{2},1;\ +\tfrac{1}{2},-1\rangle \tag{4b}$$

Add (3b) + (4b): The $|\frac{1}{2},1;\ -\frac{1}{2},0\rangle$ terms cancel:

$$\sqrt{\frac{1}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle$$
$$= \left(\frac{1}{3}+\frac{2}{3}\right)|\tfrac{1}{2},1;\ +\tfrac{1}{2},-1\rangle = 1\cdot|\tfrac{1}{2},1;\ +\tfrac{1}{2},-1\rangle \tag{33}$$

Therefore:

$$\boxed{|K^+\Sigma^-\rangle = |\tfrac{1}{2},1;\ +\tfrac{1}{2},-1\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle} \tag{Final-b}$$

Amplitude for Reaction (b)

Step 6: Compute the amplitude

Channel	Initial coeff.	Final coeff.	Product
$I = 3/2$	$\sqrt{1/3}$	$\sqrt{1/3}$	$\sqrt{1/3}\times\sqrt{1/3} = \frac{1}{3}$
$I = 1/2$	$-\sqrt{2/3}$	$+\sqrt{2/3}$	$(-\sqrt{2/3})\times(+\sqrt{2/3}) = -\frac{2}{3}$

$$\boxed{\mathcal{M}_b = \frac{1}{3}\,\mathcal{A}_{3/2} - \frac{2}{3}\,\mathcal{A}_{1/2} = \frac{1}{3}\left(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right)} \tag{34}$$

$$|\mathcal{M}_b|^2 = \frac{1}{9}\left|\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right|^2 \tag{35}$$

Reaction (c): $\pi^+ + p \to K^+ + \Sigma^+$

Step 1: Write Out the Reaction

$$\pi^+ + p \longrightarrow K^+ + \Sigma^+ \tag{36}$$

New initial state — start fresh.

Initial State: $\pi^+ + p$

Step 2: Find I and I_3, write the kets

Pion ($\pi^+ = u\bar{d}$): $I = 1, \quad I_3 = +1.$
Proton: $I = 1/2, \quad I_3 = +1/2.$
Direct-product ket:

$$|\pi^+ p\rangle = |1, \tfrac{1}{2};\ +1, +\tfrac{1}{2}\rangle \tag{37}$$

Step 3: Triangle rule and I_3 conservation

Triangle rule: $I = 1/2$ or $3/2$ (same as before).
$I_3 = (+1) + (+1/2) = +3/2$.
Check which $|J, M\rangle$ states can have $M = +3/2$:

- $|3/2, +3/2\rangle$: $I = 3/2$ allows $I_3 = +3/2$. **Yes.**
- $|1/2, +3/2\rangle$: $I = 1/2$ only goes up to $I_3 = +1/2$. **No! Doesn't fit.**

Only one ket: $|3/2, +3/2\rangle$.

Steps 4–5: Decomposition is trivial (stretched state)

Only one product state has $m_1 + m_2 = +3/2$: that's $m_1 = +1, m_2 = +1/2$.
The CG coefficient is:

$$\langle 1, \tfrac{1}{2};\ +1, +\tfrac{1}{2} \,|\, \tfrac{3}{2}, +\tfrac{3}{2}\rangle = 1 \quad \text{(table entry: 1, and } \sqrt{1} = 1) \tag{38}$$

The decomposition has only one term, so inversion is trivial:

$$\left|\frac{3}{2}, +\frac{3}{2}\right\rangle = |1, \tfrac{1}{2};\ +1, +\tfrac{1}{2}\rangle = |\pi^+ p\rangle \tag{39}$$

No system of equations needed!

$$\boxed{|\pi^+ p\rangle = |1, \tfrac{1}{2};\ +1, +\tfrac{1}{2}\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle} \tag{Initial-c}$$

Final State: $K^+ + \Sigma^+$

Step 2: Find I and I_3, write the kets

Kaon ($K^+ = u\bar{s}$): $I = 1/2, \quad I_3 = +1/2.$
Sigma ($\Sigma^+ = uus$): $I = 1, \quad I_3 = +1.$
Direct-product ket (kaon first):

$$|K^+\Sigma^+\rangle = |\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, +1\rangle \tag{40}$$

$I_3 = (+1/2) + (+1) = +3/2.$

Steps 3–5: Same stretched state argument

$I_3 = +3/2$ only fits in $I = 3/2$. Only one product state. The CG coefficient (with the coupling-order phase, which is $+1$ for $J = 3/2$) is:

$$\langle \tfrac{1}{2}, 1;\ +\tfrac{1}{2}, +1 \,|\, \tfrac{3}{2}, +\tfrac{3}{2}\rangle = (+1) \times \langle 1, \tfrac{1}{2};\ +1, +\tfrac{1}{2} \,|\, \tfrac{3}{2}, +\tfrac{3}{2}\rangle = (+1)(1) = 1 \tag{41}$$

Trivial inversion:

$$\boxed{|K^+\Sigma^+\rangle = |\tfrac{1}{2}, 1;\ +\tfrac{1}{2}, +1\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle} \tag{Final-c}$$

Amplitude for Reaction (c)

Step 6: Compute the amplitude

Both initial and final are pure $|3/2, +3/2\rangle$ with coefficient 1.

Channel	Initial coeff.	Final coeff.	Product
$I = 3/2$	1	1	$1 \times 1 = 1$
$I = 1/2$	0	0	0

$$\boxed{\mathcal{M}_c = \mathcal{A}_{3/2}} \tag{42}$$

$$|\mathcal{M}_c|^2 = |\mathcal{A}_{3/2}|^2 \tag{43}$$

Key observation

$|\pi^+ p\rangle = |1, \tfrac{1}{2};\ +1, +\tfrac{1}{2}\rangle$ is pure $I = 3/2$. It has **zero** coefficient for $I = 1/2$. So when $I = 1/2$ dominates, this reaction is completely forbidden: $\sigma_c = 0$.

Taking the Ratios

Collecting all amplitudes

$$\mathcal{M}_a = \frac{\sqrt{2}}{3}\left(\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\right) \quad \Longrightarrow \quad |\mathcal{M}_a|^2 = \frac{2}{9}\left|\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\right|^2 \tag{44}$$

$$\mathcal{M}_b = \frac{1}{3}\left(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right) \quad \Longrightarrow \quad |\mathcal{M}_b|^2 = \frac{1}{9}\left|\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right|^2 \tag{45}$$

$$\mathcal{M}_c = \mathcal{A}_{3/2} \quad \Longrightarrow \quad |\mathcal{M}_c|^2 = \left|\mathcal{A}_{3/2}\right|^2 \tag{46}$$

Cross section: $\sigma \propto |\mathcal{M}|^2$.

Case 1: $I = 3/2$ dominates (set $\mathcal{A}_{1/2} = 0$)

$$\sigma_a \propto \frac{2}{9}|\mathcal{A}_{3/2}|^2 \tag{47}$$

$$\sigma_b \propto \frac{1}{9}|\mathcal{A}_{3/2}|^2 \tag{48}$$

$$\sigma_c \propto |\mathcal{A}_{3/2}|^2 \tag{49}$$

$|\mathcal{A}_{3/2}|^2$ cancels. Multiply by 9:

$$\boxed{\sigma_a : \sigma_b : \sigma_c = 2 : 1 : 9} \tag{50}$$

Case 2: $I = 1/2$ dominates (set $\mathcal{A}_{3/2} = 0$)

$$\sigma_a \propto \frac{2}{9}|\mathcal{A}_{1/2}|^2 \tag{51}$$

$$\sigma_b \propto \frac{1}{9} \cdot 4\,|\mathcal{A}_{1/2}|^2 = \frac{4}{9}|\mathcal{A}_{1/2}|^2 \tag{52}$$

$$\sigma_c \propto 0 \tag{53}$$

$|\mathcal{A}_{1/2}|^2$ cancels. Multiply by 9/2:

$$\boxed{\sigma_a : \sigma_b : \sigma_c = 1 : 2 : 0} \tag{54}$$

Final Answer

Summary

	Amplitude $\mathcal{M}$	$I = 3/2$ dom.	$I = 1/2$ dom.
(a) $\pi^- p \to K^0\Sigma^0$	$\frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2})$	2	1
(b) $\pi^- p \to K^+\Sigma^-$	$\frac{1}{3}(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2})$	1	2
(c) $\pi^+ p \to K^+\Sigma^+$	$\mathcal{A}_{3/2}$	9	0

Consistency check

$$\mathcal{M}_c = \sqrt{2}\,\mathcal{M}_a + \mathcal{M}_b \tag{55}$$

$$\begin{aligned}\sqrt{2}\cdot\frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2}) + \frac{1}{3}(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}) &= \frac{2}{3}\mathcal{A}_{3/2} + \frac{2}{3}\mathcal{A}_{1/2} + \frac{1}{3}\mathcal{A}_{3/2} - \frac{2}{3}\mathcal{A}_{1/2} \\ &= \mathcal{A}_{3/2} = \mathcal{M}_c \quad \checkmark \end{aligned} \tag{56}$$

The complete recipe using Zettili notation

1. **Write the reaction.**
2. **Find I and I_3** for each particle. Write each as $|I, I_3\rangle$. Write the direct-product ket as $|j_1, j_2;\ m_1, m_2\rangle$.
3. **Triangle rule**: $|j_1 - j_2| \le J \le j_1 + j_2$ gives the allowed total isospins.
4. **I_3 conservation**: $M = m_1 + m_2$ tells you which $|J, M\rangle$ kets you need.
5. **Decompose each $|J, M\rangle$** using the CG table and the Zettili formula:

$$|J, M\rangle = \sum_{m_1, m_2} \langle j_1, j_2;\ m_1, m_2 \,|\, J, M\rangle\ |j_1, j_2;\ m_1, m_2\rangle$$

   If j_1, j_2 are in the wrong order for the table, swap and apply phase $(-1)^{j_1+j_2-J}$.
6. **Invert with systems of equations**: multiply equations by appropriate factors, add to cancel the unwanted product ket, and isolate the one you want. This gives the reaction's direct-product ket in terms of $|J, M\rangle$ kets.
7. **Compute amplitude**: $\mathcal{M} = \sum_I (\text{initial CG coeff.}) \times \mathcal{A}_I \times (\text{final CG coeff.})$
8. **Square and take ratios**: $\sigma \propto |\mathcal{M}|^2$.